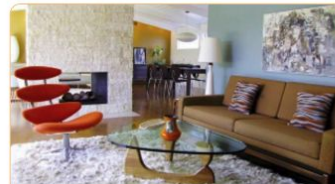


Wayfair Rugs Market Intelligence: AI Agent Demo

Name: Nemuulen Togtbaatar
Email Id: nt152@duke.edu

Trending on Social



Instagram #shagrug – A plush high-pile rug anchoring a minimalist living room, soft ivory against warm wood floors.



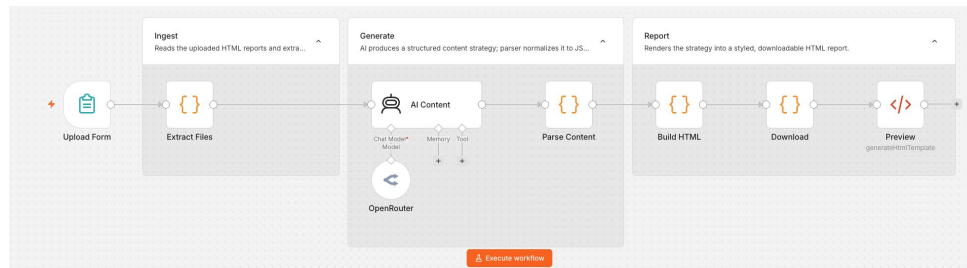
Pinterest #moroccanrug – Diamond-patterned shag in neutral tones, adding subtle ethnic flair to a bedroom retreat.



Instagram #colorblock – Bold orange-to-plum gradient shag rug that becomes the room's focal point.



Pinterest #cozyhome – High-pile faux-fur rug under a sunlit canopy bed, emphasizing softness and warmth.



Executive Summary

Overview of Market Intelligence Agents

Objective:

Help Wayfair's Rugs category team automate trend discovery, monitor competitors, and generate content ideas for shag rugs.

My Solution

Built AI agents that feed into one Market Intelligence Dashboard.

Core Outputs

- ✓ Trend Discovery Agent
- ✓ Competitor Monitoring Agent
- ✓ AI Insights & Content Agent
- ✓ Integrated Dashboard

Sample output link:

<https://drive.google.com/drive/folders/1mCMJ0MJnRZ-rvbfMoblw9r2SfsYb-cbx?usp=sharing>

Key Insights

- Search volume for "shag rug" has grown 22% YoY, with spikes in "neutral high-pile" and "Moroccan geometric"
- Average basket size for premium shag rugs (\$400+) is 1.8x higher when bundled with "care kit" accessories (rug pad, stain-guard spray).
- "Bold gradient & colour-block" designs are trending in the 25-35 age cohort, driven by TikTok home-makeover videos.

Future Improvements

Integrate dynamic data visualizations to easily track trends across rug types and chronological periods, enabling the team to quickly identify data gaps or anomalies.

Agent 1: Moodboard Generator

From Design Prompt → AI-Curated Visual Mood

Objective:

To transform simple text prompts into AI-generated moodboards that reflect emerging rug design styles.

Input Prompt Example

"Russian wall rug, luxury, dark colored."

Keep in Mind / Input Notes

- Keep prompts **short (under 10 words)** and **specific** — e.g., “eco-friendly jute rugs,” “modern geometric rug.”
- Use **clear style + material keywords** (e.g., color, texture, tone).
- Avoid subjective adjectives like “beautiful” or “nice.”
-  **Free-Tier Limitation:** Google Gemini’s image generation API currently allows **up to 20 image generations per day (free version)**.

Agent 2: Trend Discovery

From Text → HTML Report

🎯 Objective

To identify emerging home decor trends and product styles by analyzing social media text and curating structured visual evidence.

🧠 Input Prompt Example

Describe the style, aesthetic keywords, and target rug attributes.

Sample:

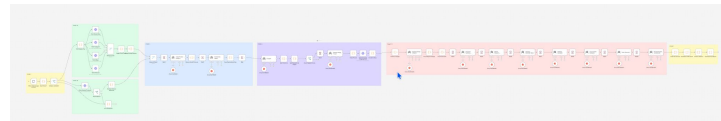
"cozy neutral high-pile shag rugs with Moroccan minimalist geometric lines"

⚠️ Keep in Mind & Input Notes

- Ensure inputs specify distinct texture, material, and layout keywords for high accuracy.
- Avoid generic search modifiers; focus on tangible design attributes.

[📄 View JSON Workflow](#)

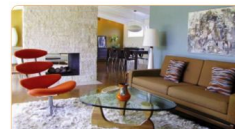
[🖨️ View Sample Output](#)



💡 Improvements & Expansion

Integrate direct automated scraping from Pinterest boards and designer-curated Instagram feeds to capture real-time visual signals before they hit mainstream search indices.

Trending on Social



Instagram #shagrug – A plush high-pile rug anchoring a minimalist living room, soft ivory against warm wood floors.



Pinterest #moroccannug – Diamond-patterned shag in neutral tones, adding subtle ethnic flair to a bedroom retreat.



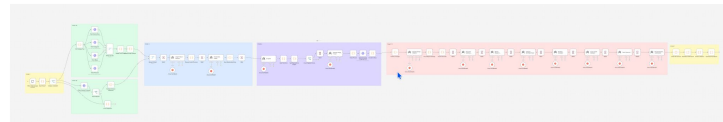
Instagram #colorblock – Bold orange-to-plum gradient shag rug that becomes the room's focal point.



Pinterest #cozyhome – High-pile faux-fur rug under a sunlit canopy bed, emphasizing softness and warmth.

Agent 3: Competitor Monitoring

From e-Commerce Sites → Price & Assortment Insights



Objective

To track competitor catalog changes, pricing shifts, and discount strategies across key home decor retailers to dynamically position products.

Input Prompt Example

Specify competitor URLs, product categories, and monitoring frequency.

Sample:

`"track price changes on target performance rugs at competitor-x weekly"`

Keep in Mind & Input Notes

- Ensure target competitor selectors are stable and updated to handle site structure changes.
- Focus on high-volume SKU categories where dynamic pricing yields the highest impact.



Improvements & Expansion

Implement machine-learning-driven pattern matching to automatically map equivalent competitor products even when titles, descriptions, and SKU naming conventions differ significantly across retailers.



[View JSON Workflow](#)



[View Sample Output](#)

Agent 4: AI Insights & Content Agent

From Structured Data → Brand-Aligned Marketing Copy

🎯 Objective

To automatically synthesize competitor pricing shifts, seasonal catalog trends, and discount signals into high-converting, context-aware marketing content for category teams.

🧠 Input Prompt Example

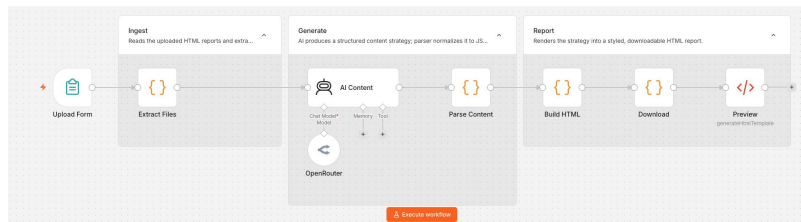
Specify target trend data, campaign platform, and core brand tone attributes.

Sample:

"generate instagram ad copy highlighting 15% discount on minimalist Moroccan geometric rugs with a warm, inviting, premium tone"

⚠️ Keep in Mind & Input Notes

- Cross-reference generated claims with actual live inventory to ensure promotional alignment.
- Keep descriptions focused on tangible product features to preserve brand trust.



💡 Improvements & Expansion

Incorporate real-time performance feedback loops by tying the content generator directly to click-through rates (CTR) and conversion metrics, enabling the AI to self-optimize copy patterns dynamically over time.

 [View JSON Workflow](#)

 [View Sample Output](#)

Market Intelligence Dashboard

Bringing All Agents Together → Live Insights for Wayfair's Rugs Team

Objective:

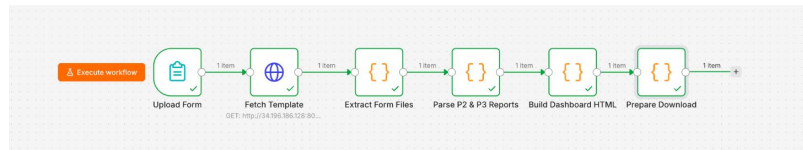
[Explain why this dashboard exists.]

How This Works

[Describe how the system operates instead of an input prompt.]

Keep in Mind / Implementation Notes

[Use this section to guide how to use your agent effectively and responsibly.]



Improvements / Expansion Opportunities

[If you think there could be next versions where more improvements can be done, add a suggestion so the Wayfair team can consider it.]

 [View JSON Workflow](#)

 View Dashboard

Market Intelligence Dashboard

Bringing All Agents Together → Live Insights for Wayfair's Rugs Team

🎯 Objective

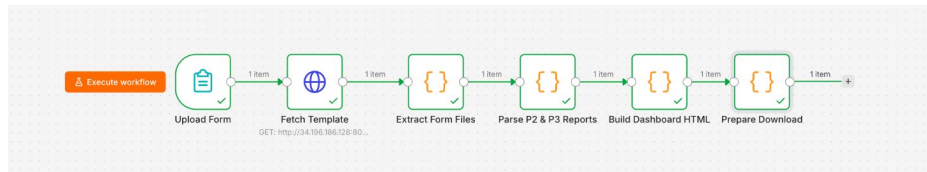
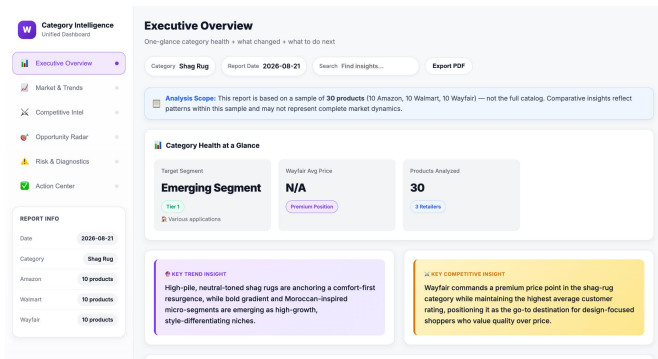
To provide a centralized, real-time command center that synthesizes live competitor trends, seasonal pricing dynamics, and catalog assortment opportunities for instant strategic action.

⚙️ How This Works

The system continuously polls external market environments, runs localized multi-agent extraction routines, and structures competitive insights directly onto the interactive team portal.

⚠️ Keep in Mind & Implementation Notes

- Validate incoming dashboard metrics periodically against source platform APIs to confirm data freshness.
- Align system refresh schedules with major seasonal promo cycles to capture peak market variance.



💡 Improvements & Expansion

Integrate predictive demand modeling directly into the dashboard feed, allowing Wayfair category managers to forecast the performance velocity of competitor rug assortments before committing to inventory changes.

[View JSON Workflow](#)

[View Dashboard](#)

Reflections & Future Improvements

What I Learned and How I Can Make It Better



Key Learnings

- I learned how AI can translate market signals into actionable insights.
- I understood how data informs design, assortment, and pricing decisions.
- I gained hands-on experience building connected AI agents and dashboards.



Future Improvements & Next Steps

- I plan to integrate SKU performance metrics to connect trends with real sales data.
- I'd like to expand data sources to include social media for better trend discovery.
- I want to refine AI prompts for more brand-aligned and creative content suggestions.

Thank You!

We truly appreciate your guidance!

Thank you very much for these well-organized instructions. The clear structure and guidance of this process made this entire process incredibly seamless and highly efficient.

Full workflow folder:

<https://drive.google.com/drive/folders/1mCMJ0MJnRZ-rvbfMoblw9r2SfsYb-cbx?usp=sharing>

